

Westravelle Analytical Ltd, a Tarnovell company

AR-27: approved analytical report

Assay and related-substances validation

Report no. E-R-2020-224-01

Project: WA/2020/224 | LIMS: WST-20-006965

Client: Ardelvune Pharmaceuticals

Protocol interval: 24 Aug 2020 to 06 Oct 2020

Report date: 08 Oct 2020

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Status: Final, QA-approved

1 Summary and principal evidence

The robustness protocol identified sensitivity to mobile-phase preparation and Ardelvune tightened the buffer instructions.

The following tables preserve the approved numerical data, units, result qualifiers and acceptance entries. They are read with batch-card identity, preparation and system suitability information. The scientific interpretation and client action are developed in the numbered sections.

Table 1. Approved validation summary; units and qualifiers retained.

Parameter	Result	Acceptance
Linearity (r^2)	0.9992	≥ 0.999
Range (% nominal)	50-150	80-120 minimum
Accuracy 80% (% recovery)	100.5	98.0-102.0
Accuracy 100% (% recovery)	99.9	98.0-102.0
Accuracy 120% (% recovery)	100.3	98.0-102.0
Repeatability (%RSD, n=6)	0.30	≤ 2.0
Intermediate precision (%RSD)	0.81	≤ 2.0
LOQ (% of nominal)	0.05	≤ 0.05

2 Report authority

This approved report is addressed to Ardelvune Pharmaceuticals. Alice Penrose is responsible for scientific interpretation and release. Leon Wu maintains the sample and storage links; Saira Mahmood maintains the analytical method and system suitability links. The accession WST-20-006965 connects analytical preparations to batch cards, without replacing the client stability lot identity.

The signed report is the reporting copy. Sequence sheets, storage logs and original chromatographic files remain supporting material. An extract must retain the condition, timepoint, units and result qualifiers. Client decisions are distinguished from analytical findings in the conclusion.

3 Protocol objective

The work concerns AR-27 supplied for Ardelvune Pharmaceuticals, under WA/2020/224. Testing opened on 24 Aug 2020, the interim review was dated 14 Sep 2020, and analytical completion was recorded on 06 Oct 2020. The objective was to establish the analytical basis for the stated client decision.

Source material, exposure and preparation route belong to the result. A difference between analytical preparations is not automatically a difference between formulations. Investigative observations are discussed separately from accepted tabulated results. No extrapolated result is inserted into the reporting table.

4 Source-material traceability

The source chain begins with receipt and client stability lot description, continues through the batch card and withdrawal, and ends with the sequence sheet and approved result. Each preparation is linked to its source container. WST-20-006965 is the accession anchor for this chain.

The report does not create alternative stability lot numbers as free-text aliases. The source identifier remains on the batch card. A storage-condition label alone is insufficient evidence of sample identity. Source identity was reconciled before analytical trends were considered.

5 Receipt and custody

Product description, container state and accompanying instructions were entered against WST-20-006965. The custody trail distinguishes material reserved for the protocol from material removed for analysis. Handling after withdrawal is recorded because bench exposure can affect interpretation of a humidity comparison.

Retained material remains identifiable to its source container. A preparation remainder is not returned to that container. An investigative retained sample requires its exposure history to be established before comparison with scheduled material. The report assigns no unrecorded exposure history to a result.

6 Batch-set allocation

The batch set was managed so that source material, analytical preparations and reference preparations remained distinct. Client stability lot identities are carried on the batch cards. The sequence identifies the preparation used for each injection and its applicable exposure or challenge condition.

A fresh preparation examines extraction or handling, whereas reinjection examines an existing solution. These operations answer different questions and are not pooled merely because they share an accession. Subsamples from a container are not counted as independently manufactured stability lots.

7 Analytical responsibilities

Alice Penrose reviewed interpretation against the protocol objective. Leon Wu reconciled storage and withdrawal information. Saira Mahmood checked analytical sequences and system suitability disposition. Mark Campbell reviewed report-to-source links for the approved copy.

Approval does not replace contemporaneous analyst entries. Corrections remain attributable on their sequence sheets and original analytical files remain available. Scientific review asks whether the evidence supports the conclusion; quality review establishes faithful reporting and traceability without creating missing numerical evidence.

8 Specificity

Specificity was considered for assay and related substances, rather than inferred from the main peak alone. Blank, reference and test preparations were compared within the sequence. Interference was assessed for the intended measurement.

Retention order does not establish chemical identity. Specificity concerns separation from interfering responses under tested conditions; it does not characterise every possible future degradation product.

9 Robustness design

The robustness assessment deliberately varied method inputs to determine which instructions needed tighter control. Mobile-phase preparation mattered because preparation order and buffer adjustment can alter chromatography without changing the named reagents.

Variations were controlled protocol conditions, not undocumented departures. Their preparation links remain on sequence sheets. A deliberate variation is not pooled with the nominal condition as an equivalent replicate.

10 Buffer-preparation investigation

Sensitivity was identified in mobile-phase preparation. The preparation route, including when the buffer condition was established relative to solvent addition, was an operational concern. The response was to make the buffer instructions explicit.

Nominal composition alone is insufficient where preparation routes alter chromatography. Tightened instructions belong in the controlled method rather than solely in correspondence or this report.

11 Assay preparation

Test solutions were prepared from identified source material using the protocol extraction and dilution route. Weighing, transfer, dissolution and dilution entries remain on the sequence sheet.

Preparations were protected from avoidable evaporation and contamination while awaiting injection.

Assay is expressed as percent label claim. The reference basis and dilution factors belong to the calculation. The report does not silently convert assay into a dry-weight result or correct it for water. Reported values are those in the approved numerical table.

12 Related-substances measurement

Related substances are reported as area percent under the applicable chromatographic method. Integration settings, peak assignment and approved qualifiers accompany the analytical sequence. Retention position assists comparison but is not, by itself, proof of chemical identity.

An investigative reprocessing remains linked to its reason and does not overwrite original integration history. A less-than entry is not replaced by zero. Individual impurities and total impurities are read in their stated units rather than treated as an assay deficit.

13 Reference-material control

Reference preparations are identified separately from test preparations. Their batch cards retain the material basis applicable to the calculation. Preparation and use are linked to the analytical sequence, so a visible reference peak is not alone evidence of a valid calibration.

A reference used for peak assignment has a different role from one used for assay calibration. An investigative addition to a sample is not automatically its quantitative result. Reference-only, sample-only and combined preparations are interpreted according to the question being investigated.

14 Sequence design

The sequence sheet separates blanks, reference, system suitability and test preparations. Injection order is retained so a sample can be related to its surrounding controls. Sequence completion does not itself establish analytical acceptance.

Blanks support assessment of interference and carry-over; bracketing reference preparations support response review. An unexpected signal is assessed against preparation history and injection position. A repeat sequence retains its own identity and link to the original. Rejected measurements are not averaged into accepted results.

15 System suitability

System suitability is a release gate for an analytical sequence. Method-defined checks are applied before sample calculations are accepted. Reference response and the separation relevant to impurity reporting are reviewed for the actual analytical question.

Detailed acceptance entries remain on the sequence sheets. This report does not invent numerical suitability values absent from the consolidated source. A sequence requiring investigation retains its initial disposition. Assay suitability cannot be presumed to establish suitability for a more demanding impurity assignment.

16 Calculations

Assay retains the percent label claim basis and related substances retain the area percent basis. Water, where tabulated, has its own mass basis. These quantities cannot be added or subtracted as interchangeable measurements. Rounding follows the controlled source table.

Missing measurements are not zero results and row status is not a numerical value. Comparisons require matched conditions and source links. The tables are reproduced from controlled project data, preserving qualifiers without manually introducing extra precision or derived result values.

17 Validation interpretation

The validation table retains each parameter, result and acceptance entry. A favourable parameter does not resolve sensitivity identified in robustness. Qualification must be read with the preparation controls.

Routine use should follow the tightened buffer instructions. This report does not extend qualification to an alternative buffer recipe, changed preparation route or different analytical scope.

18 Method implementation

Ardevune Pharmaceuticals tightened the buffer instructions after the robustness findings. Saira Mahmood retained the link between those instructions and the method qualification evidence.

Transfer should preserve preparation order and the stated buffer controls. The report does not invent a new numeric buffer recipe or claim later routine performance from the completed qualification alone.

19 Data handling

Original chromatographic files, preparation entries and sequence sheets remain associated with WST-20-006965. The archive retains injection order, processing history and the relationship between a preparation and its reported result. A revised integration is distinguishable from the original.

Report assembly uses the approved project tables rather than a later working export. Investigative records are retained even when no reported result changes, because they explain the accepted interpretation. Additional analysis is treated as additional work rather than an undocumented alteration to this final report.

20 Interim and final review

The interim review on 14 Sep 2020 was considered against the evidence then available. The final interpretation incorporates work completed by 06 Oct 2020. An observation, an investigated explanation and a client decision remain distinguishable.

Completion closes the reporting interval under WA/2020/224. It does not extend exposure or create a shelf-life claim. A receiving laboratory needs the preparation basis and suitability requirements from the controlled method rather than having to infer executable instructions from this report.

21 Conclusion

The robustness protocol identified sensitivity to mobile-phase preparation and Ardelvune Pharmaceuticals tightened the buffer instructions. Assay and related-substances validation is reported with that operational requirement.

The report closes WA/2020/224 under WST-20-006965. Controlled buffer instructions accompany the method; source preparations and sequence sheets retain the evidence supporting this outcome.

22 Archive index

The reporting pair comprises this final report and the approved method-validation sequence sheets. Analytical completion was 06 Oct 2020; source records remain under WA/2020/224, accession WST-20-006965. Batch cards, exposure records and sequence sheets remain the authoritative detailed links.

The original project-year identifier is retained even where completion occurred in a later year. Branding follows the report date. A later commercial identity is not applied retrospectively to the reporting copy.

23 Batch-card retrieval

Retrieve the batch card using WST-20-006965, then confirm client stability lot and container before retrieving the sequence. Searching only by product name can return material from another exposure or preparation.

A result query should preserve the reported condition, timepoint and sequence identity. That trail distinguishes a material question from a processing question. This appendix is a retrieval guide, not a replacement batch card.

24 Sequence-sheet retrieval

The sequence sheet retains preparation entries, injection order, references and system suitability disposition. Retrieve the accepted original before comparing a reported value with an investigative reinjection.

Where reprocessing was used, retain original and revised processing links and the reason. A later working export cannot alone determine which preparation supplied the approved result. Qualifiers remain in any query response.

25 Result-query routing

Specify the report identifier, accession and affected row. Establish whether the concern is identity, exposure, preparation, integration, calculation or interpretation. Different questions require different records.

Alice Penrose handles interpretation, Leon Wu storage and material links, and Saira Mahmood method and sequence links. A response should preserve original units and limitations. A query does not authorize overwriting an approved result.

26 Controlled vocabulary

Protocol denotes the agreed programme; batch set the material grouping; sequence sheet the preparation and injection record; system suitability the analytical acceptance gate. Stability lot is source identity, not every new preparation.

A deviation note retains the issue and disposition. A release gate is an evidence-based decision point. Consistent vocabulary prevents storage, analytical acceptance and client action from being conflated in subsequent extracts.

27 Units and qualifiers

Assay is percent label claim and related substances area percent. Chamber temperature, where applicable, uses °C. Water KF retains the table's percent w/w basis. Headings belong to the result.

A qualified low-level entry is not an exact zero. Status remains with the row. No derived value from rounded data, retrospective water correction or new acceptance limit is introduced in this report.

28 Handover record

The handover retains the approved protocol, method, numerical tables and client action under WA/2020/224. It should also retain preparation and reference instructions needed to interpret the findings.

Transfer of a report is not proof of receiving-site performance. Additional work needs its own source links, objective and acceptance disposition. The conclusion here stays within the completed interval.

Name	Role	Signature	Date
Alice Penrose	Head of Analytical Services — scientific approval	<u>A Penrose</u>	08 Oct 2020
Leon Wu	Stability Coordinator — material traceability	<u>L Wu</u>	08 Oct 2020
Saira Mahmood	QC Method Validation Lead — analytical review	<u>S Mahmood</u>	08 Oct 2020
Mark Campbell	QA Officer — reporting-copy approval	<u>M Campbell</u>	08 Oct 2020